Guarded solving degree and regularity

Rigorous degree bounds and explicit presentation controls — 30 October 2026

The prior scoped Lean theorem proves the exact ordinary first/last pair $(4, 5)$ for the guarded three-abscissa prime-field presentation. This follow-up separates last fall, closed-Macaulay solving degree and the Bardet–Faugère–Salvy (BFS) degree of regularity. No exhaustive campaign or performance measurement is reported.

Actual inputs. Work in $k[u, x, y, z, v]$ with $g(T) = (T - a)(T - b)(T - c)$ and inputs

$$g(x),\ g(y),\ g(z),\ S_3(x, y, u),\ S_3(u, z, r),\ v(x - y) - 1.$$

For ordinary binary curves use $S_3(s, t, w) = (st + sw + tw)^2 + stw + a_6$; the guard in characteristic two is $v(x + y) + 1$. There is no implicit Boolean reduction or field equation.

Sharp solving degree. The original prime hypotheses are retained: prime $p > 3$, distinct base abscissas, nonsingular curve, rational target and base lifts, and polynomial NoSolution. The prior proof constructs $1 \in V(F, 5)$ and excludes it below five. For a unit ideal, a Gröbner basis in $V(F, d)$ must contain a nonzero constant, by its leading-monomial criterion applied to 1. Conversely $\{1\}$ is a Gröbner basis. Lean checks this equivalence and the exact closed solving degree five for every monomial order. For the classical degree-compatible solving degree, Caminata–Gorla Theorem 3.1 gives $\mathrm{sd} = \max(d_F, 0) = 5$. The general bridge to a particular matrix algorithm is a cited written corollary, not newly formalized software.

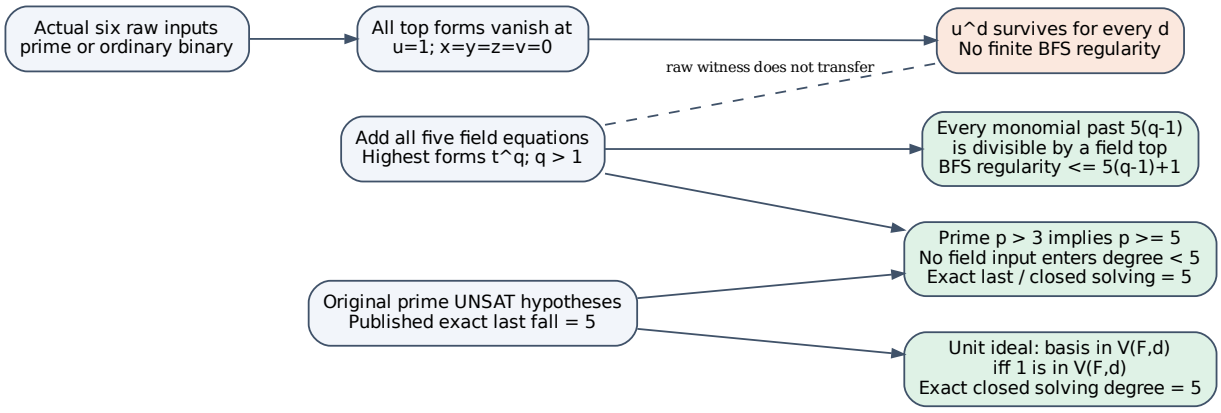
No finite raw BFS regularity. All six actual top forms vanish at $(u, x, y, z, v) = (1, 0, 0, 0, 0)$, in both the prime and binary formulas. Yet u^d evaluates to 1 for every $d \geq 0$. Thus no homogeneous piece of the top ideal is full. The finiteness assumption of BFS regularity fails. This says nothing about infinite Castelnuovo–Mumford regularity.

Explicit field-equation control. Adjoining $t^q - t$ for all five variables adds actual top forms t^q for $q > 1$. The previous witness belongs to the augmented top ideal once $d \geq q$. Every monomial of degree greater than $5(q - 1)$ is divisible by a field top. Lean proves

$$d_{\mathrm{reg}} \leq 5(q - 1) + 1.$$

For primes $p > 3$, those field equations cannot enter a derivation below five; the exact last fall and closed solving degree remain five after adjoining them. The unit ideal is constructed, not assumed.

Presentation	Last fall	Closed solving	BFS regularity
Raw prime, original hypotheses	5	5	No finite value
Prime plus field equations	5	5	$\leq 5(p - 1) + 1$
Raw ordinary binary	Not proved here	Not proved here	No finite value
Binary plus field equations	Not proved here	Not proved here	$\leq 5(q - 1) + 1$



Proof dependency diagram. All displayed bounds are mathematical statements, not experimental measurements.

Formal coverage. The added module is `DegreeInvariants.lean`; its proof map distinguishes explicit closure/basis definitions from the cited general classical solving-degree bridge. Source/configuration hashes and actual default build, transitive axiom audit and fresh kernel replay outcomes are retained separately in `validation/`. Earlier theorem sources, hypotheses and frozen proof-study evidence are unchanged. Failed development attempts remain recorded separately. The complete repository library check passed 5,430 tests, zero failed, 123 ignored; native draft-card validation passed after exact tracked schema restoration, with the original setup failure retained.

Exhaustive study remains open. The designed census uses $y^2 = x^3 + 1$ over $\mathbb{F}_7$ and $y^2 + xy = x^3 + x^2 + 1$ over $\mathbb{F}_8 = \mathbb{F}_2[t]/(t^3 + t + 1)$. It includes every affine target and every three-element rational-abscissa base, with and without explicit field equations. The bounds 31 and 36 for the two augmented top ideals are proved upper bounds, not measured exact degrees.

Positive, planted-SAT, representation, A/A-permutation and changed-payload corruption controls are specified. Complete upper/lower, basis and Hilbert certificates are required; SAT cases cannot be assigned a last fall from a non-contradiction or stabilization flag. No native census instrument or canonical frozen protocol is claimed complete. AWS STS currently reports missing credentials, and the reviewed GitHub route builds ecbench rather than the intended instrument. No scientific run, integration-fixture substitute or unauthorized local sweep was performed. The blocked draft backlog card retains these obligations and has zero funding.

Sources checked before implementation. Caminata–Gorla, arXiv:2112.05579v2, Definitions 1.1/1.4/1.5 and Theorem 3.1; Semaev, arXiv:1504.01175, Section 2; the existing ledger and exact-pair/top/closure Lean sources. No novelty, asymptotic, recovery or speed claim is made.